

AZ-900 Mock Practice Quiz

Questions, options, correct answer, and explanation. Practice/mock material - not an official Microsoft exam dump.

Q1. To transition a large capital expenditure (CapEx) to an operational expenditure (OpEx), Contoso's CIO has suggested to the board that the organization move their virtual machine (IaaS) workloads to Azure. The CFO argues that this will result in unpredictable OpEx spending. In this case, what pricing option is available to reduce costs and making predicting future spending easier?

- A. Azure Reservations
- B. Pay-as-you-go (PAYG)
- C. Azure VM Scale Sets
- D. Azure in CSP

Right option: A. Azure Reservations

Explanation: Azure Reservations are designed for predictable, long-running workloads. By committing to use resources for a fixed term, Contoso can reduce VM costs and make future spending easier to forecast than pure pay-as-you-go.

Remember: Predictable long-running workloads = Reservations.

Q2. Tailspin Toys, a small startup, is choosing their initial services architecture. They choose to rely on Office 365 and Microsoft Azure for all services. Which of the following best describes the cloud model they have chosen?

- A. Private Cloud
- B. Public Cloud
- C. Hybrid Cloud
- D. Government Cloud

Right option: B. Public Cloud

Explanation: Office 365 and Azure are Microsoft-operated cloud services available over the public internet. Tailspin is consuming services from a public cloud provider rather than running a private datacenter.

Remember: Office 365 and Azure as hosted Microsoft services = public cloud.

Q3. Which of the following statements are true of a Software as a Service solution?

- A. You are responsible for maintaining the solution infrastructure
- B. You are responsible for deploying updates to the solution
- C. You are responsible for solution availability and scalability
- D. You are responsible for configuring the solution features

Right option: D. You are responsible for configuring the solution features

Explanation: In SaaS, the provider manages the application infrastructure, updates, availability, and scaling. The customer usually configures users, settings, and features of the service.

Remember: In SaaS, the customer configures features; the provider runs the service.

Q4. Contoso is planning to move to Azure, but currently hosts business applications in a shared virtualization infrastructure on-premises, utilizing Hyper-V. This is an example of which cloud computing model?

- A. Hybrid Cloud
- B. Private Cloud
- C. Public Cloud
- D. Government Cloud

Right option: B. Private Cloud

Explanation: A private cloud is cloud-like infrastructure dedicated to one organization. Because the virtualization platform is on-premises and controlled by Contoso, it is private cloud rather than hybrid cloud.

Remember: On-premises cloud-like infrastructure for one organization = private cloud.

Q5. Azure SQL Database and an on-premises SQL cluster represent a(n) ___ expense and a(n) ___ expense, respectively.

- A. Capital, Capital
- B. Capital, Operational
- C. Operational, Operational
- D. Operational, Capital

Right option: D. Operational, Capital

Explanation: Azure SQL Database is paid as an ongoing cloud service, so it is operational expense. Buying and maintaining an on-premises SQL cluster requires hardware investment, so it is capital expense.

Remember: Cloud services are OpEx; owned hardware is CapEx.

Q6. In Azure, you are charged only for what you use. This is known as a

- A. Fixed-price model
- B. Consumption based model

Right option: B. Consumption based model

Explanation: A consumption-based model means you pay only for the resources you actually use. This is the standard cloud pricing idea behind pay-as-you-go services.

Remember: Consumption-based means pay only for actual usage.

Q7. Contoso runs several business applications in Azure VMs. Which cloud computing model best categorizes Azure VMs?

- A. Software as a service (SaaS)
- B. Platform as a service (PaaS)
- C. Infrastructure as a service (IaaS)
- D. Function as a service (FaaS)

Right option: C. Infrastructure as a service (IaaS)

Explanation: Azure VMs provide virtualized compute infrastructure, but you still manage the operating system, patches, and applications. That makes VMs Infrastructure as a Service.

Remember: VMs mean you manage the OS: IaaS.

Q8. Azure App Service, Azure SQL Database, and Cosmos DB are examples of which category of cloud computing service?

- A. Function as a Service (FaaS)
- B. Software as a Service (SaaS)
- C. Platform as a Service (PaaS)
- D. Infrastructure as a Service (IaaS)

Right option: C. Platform as a Service (PaaS)

Explanation: App Service, Azure SQL Database, and Cosmos DB are managed platforms. Azure handles much of the underlying infrastructure, so the customer focuses on code, data, and configuration.

Remember: Managed app and database platforms are PaaS.

Q9. You need to provide additional capacity than what is currently available in your on-premises datacenter. The solution must minimize capital expense (CapEx) and operational expense (OpEx). Which solution should you recommend?

- A. Migration to public cloud
- B. A private cloud
- C. A hybrid cloud
- D. Additional on-premises servers

Right option: C. A hybrid cloud

Explanation: A hybrid cloud lets Contoso extend on-premises capacity into Azure without immediately buying more datacenter hardware. It combines existing private/on-premises resources with public cloud capacity.

Remember: Hybrid cloud extends on-premises with public cloud.

Q10. Office 365, Azure VMs, and Event Grid represent, _____, _____, and _____, respectively.

- A. 1) SaaS 2) PaaS 3) IaaS
- B. 1) PaaS 2) IaaS 3) SaaS
- C. 1) SaaS 2) IaaS 3) PaaS

Right option: C. 1) SaaS 2) IaaS 3) PaaS

Explanation: Office 365 is SaaS because Microsoft runs the full application. Azure VMs are IaaS because you manage the OS. Event Grid is a managed platform/eventing service, so it fits PaaS.

Remember: SaaS, IaaS, and PaaS depend on who manages the stack.

Q11. Contoso hosts databases for customer-facing web applications in Azure MySQL Database. Which cloud computing model best categorizes this service?

- A. Software as a service (SaaS)
- B. Platform as a service (PaaS)
- C. Infrastructure as a service (IaaS)
- D. Function as a service (FaaS)

Right option: B. Platform as a service (PaaS)

Explanation: Azure Database for MySQL is a managed database platform. Microsoft manages the database service infrastructure, so the customer does not manage the underlying servers.

Remember: Managed databases usually map to PaaS.

Q12. Contoso hosts a legacy CRM application that runs on an Azure VM scale set. The application runs at a steady state of around 30% resource utilization. However, during month end closing activities, the application spikes to 90% resource utilization for extended periods during the last week of the month. Azure allows Contoso IT to meet these spikes in resource demand at the push of a button, paying for the additional capacity only when needed. This is an example of:

- A. Scalability
- B. Elasticity
- C. Fault Tolerance
- D. High Availability

Right option: B. Elasticity

Explanation: Elasticity is the ability to increase and decrease capacity as demand changes. The clue is paying for extra capacity only during the month-end spike.

Remember: Elasticity expands and shrinks with demand.

Q13. Contoso plans to migrate their existing on-premises SQL VMs to Azure. Which expenditure model does this implement?

- A. Capital (CAPEX)
- B. Operational (OPEX)
- C. Subscription
- D. Elastic

Right option: B. Operational (OPEX)

Explanation: Moving workloads from owned on-premises servers to Azure changes spending toward subscription or usage-based operational expense instead of upfront capital purchases.

Remember: Cloud migration usually shifts CapEx to OpEx.

Q14. A financial analysis of migration of on-premises files to Azure Storage at Contoso showed that storing 2TB of archived data in Azure would be less expensive than hosting this data on-premises. The report also showed unit cost in Azure would decrease even further as the data archive grows. This is an example of cloud

- A. Elasticity
- B. Economies of scale
- C. Scalability
- D. High Availability

Right option: B. Economies of scale

Explanation: Economies of scale mean the cloud provider can operate at huge scale and pass lower unit costs to customers. As the archive grows, the per-unit storage economics can improve.

Remember: Large cloud providers create economies of scale.

Q15. The Contoso Corp Financial Services team needs to automate several business processes. They need to create workflows in a low-code environment with a visual interface. The service must have built-in connectivity to their existing platforms, like Salesforce and SAP. The members working on the project are only lightly technical (citizen developers). Which Microsoft service will best fit their use case?

- A. Power Automate
- B. Logic Apps
- C. Azure Automation
- D. Azure Functions

Right option: A. Power Automate

Explanation: Power Automate is aimed at low-code workflow automation for business users and citizen developers. It includes many connectors for services such as Salesforce and SAP.

Remember: Low-code business workflows = Power Automate.

Q16. Kelly, who works in IT Operations, wants to automate a task using a script she wrote. She wants to do this with a minimum of expense and maintenance effort. She selected Azure Functions to host the job, instead of Azure Virtual Machines (IaaS). Does this service meet the solution criteria?

- A. YES
- B. NO

Right option: A. YES

Explanation: Azure Functions can run scripts or code without managing servers. It usually has less administration and can be cheaper for intermittent or on-demand jobs than running a VM continuously.

Remember: Functions fits short code jobs without servers.

Q17. Contoso has a line-of-business application that requires access to a file share. You need to host this share in Azure with a minimum of cost and administration effort. Which service would best fulfill this requirement?

- A. Azure Files
- B. Azure Blob Storage
- C. Azure VMs
- D. VM Scale Sets

Right option: A. Azure Files

Explanation: Azure Files provides managed file shares that applications can access like traditional file shares. It avoids the need to build and maintain a file server VM.

Remember: Managed file shares = Azure Files.

Q18. Contoso IT needs to ensure deployments of like Azure resources are the same for every deployment. Which of the following could be used to automate resource deployment?

- A. Azure API Management
- B. Management Groups
- C. Azure Resource Manager (ARM) templates
- D. Azure Synapse

Right option: C. Azure Resource Manager (ARM) templates

Explanation: ARM templates define Azure resources as code. They help deploy the same resource configuration repeatedly and consistently.

Remember: Repeatable resource deployment = ARM templates.

Q19. You are testing new software in an Azure VM. When you are done testing, you shut down the VM, which shows a state of "Stopped" in the Azure portal. Will you incur additional costs while the VM is in this state?

- A. YES
- B. NO

Right option: A. YES

Explanation: A VM shown as Stopped can still have billable resources attached, such as disks and possibly reserved networking resources. Stopping is not the same as deleting all resources.

Remember: Stopped VMs may still leave billable resources.

Q20. You are deploying multiple instances of a custom Contoso web application. The application instances share a common management lifecycle, but will be located in different Azure regions. Can you deploy resources across multiple Azure regions in a single resource group?

- A. YES
- B. NO

Right option: A. YES

Explanation: A resource group is a logical management container. It can contain resources deployed in different Azure regions, even though the resource group itself has a metadata location.

Remember: Resource groups can hold resources from different regions.

Q21. You are responsible for recommending infrastructure architectures for applications at Contoso. Which solution would you recommend for on-demand execution of automated tasks in Python for minimum of expense?

- A. Azure Monitor
- B. Azure Cloud Shell
- C. Azure Automation
- D. Azure Functions

Right option: D. Azure Functions

Explanation: Azure Functions is built for event-driven or on-demand code execution, including Python. It avoids maintaining a server and is a good fit for low-cost automated tasks.

Remember: Event-driven Python code without servers = Functions.

Q22. Which Azure service would you use to correlate events from multiple Azure resources in a central repository? (choose the best answer)

- A. Azure Data Lake
- B. Azure Log Analytics
- C. Azure Event Grid
- D. Azure Event Hub

Right option: B. Azure Log Analytics

Explanation: Log Analytics stores and queries logs from multiple Azure resources. It is the central place to correlate and search collected operational data.

Remember: Query operational logs = Log Analytics.

Q23. An _____ protects against datacenter-level failures.

- A. Availability Set
- B. Availability Zone
- C. VM scale set
- D. Azure VM

Right option: B. Availability Zone

Explanation: An availability zone is a physically separate datacenter location inside an Azure region. Spreading resources across zones helps protect against a datacenter-level failure.

Remember: Zone = separate datacenter location in a region.

Q24. Contoso has messages from a variety of sources (many Azure services) that need to be relayed to an application. Which Azure service would be best suited to the task?

- A. Azure Functions
- B. Azure Event Grid
- C. Azure IoT Hub
- D. Azure Service Bus

Right option: B. Azure Event Grid

Explanation: Event Grid routes events from Azure services and other sources to event handlers. It is designed for event-driven messaging rather than long-running queues or IoT device management.

Remember: Event Grid routes events to handlers.

Q25. You can monitor health and availability of your Azure Kubernetes Service (AKS) cluster with:

- A. Azure Monitor
- B. Azure App Insights
- C. Microsoft Sentinel
- D. Microsoft Defender for Cloud

Right option: A. Azure Monitor

Explanation: Azure Monitor is the main Azure monitoring platform. It collects metrics and logs for resources such as AKS clusters so you can track health and availability.

Remember: Azure Monitor collects metrics and logs for resources.

Q26. Which storage tier in Azure Storage delivers the highest cost of data storage?

- A. Hot
- B. Cool
- C. Cold
- D. Archive

Right option: A. Hot

Explanation: The Hot tier is optimized for frequently accessed data, so its storage price is higher than Cool, Cold, or Archive. Lower tiers reduce storage cost but increase access or retrieval tradeoffs.

Remember: Frequently accessed data = Hot tier.

Q27. To retire some aging on-premises servers, Contoso is planning to move several SQL databases to Azure SQL Database. The CIO has asked for a tool to estimate the cost of hosting these resources in Azure. Sally suggests using the Azure Pricing Calculator to estimate costs prior to migration. Does Sally's solution meet the solution criteria?

- A. YES
- B. NO

Right option: A. YES

Explanation: The Azure Pricing Calculator is used before deployment or migration to estimate expected Azure costs. It helps model services, regions, sizes, and usage assumptions.

Remember: Estimate before deployment = Pricing Calculator.

Q28. Which storage tier in Azure Storage delivers the lowest cost of data storage?

- A. Hot
- B. Cool
- C. Cold
- D. Archive

Right option: D. Archive

Explanation: Archive storage has the lowest storage cost because it is intended for rarely accessed data. The tradeoff is higher retrieval cost and longer rehydration/access time.

Remember: Rarely accessed lowest storage cost = Archive tier.

Q29. You have a mobile application that exchanges large numbers of messages with customer devices. Which type of Azure storage is optimized for storing large numbers of messages, accessible from anywhere via authenticated HTTP or HTTPS calls?

- A. Blob Storage
- B. Queue Storage
- C. Table Storage
- D. File Storage

Right option: B. Queue Storage

Explanation: Queue Storage stores large numbers of messages for application components to process asynchronously. Messages can be accessed over authenticated HTTP or HTTPS.

Remember: App messages between components = Queue Storage.

Q30. Contoso's Security team wants to apply policy-based configuration across deployments in multiple Azure subscriptions. Which of the following can help achieve this requirement?

- A. Resource Groups
- B. Management Groups
- C. Role Based Access Control (RBAC)
- D. Access Policies

Right option: B. Management Groups

Explanation: Management groups organize multiple subscriptions into a hierarchy. Policies and governance settings can then be applied across many subscriptions at once.

Remember: Subscription hierarchy and governance = Management Groups.

Q31. Contoso plans to implement a hybrid cloud architecture utilizing Azure. They need to connect on-premises application resources. The solution should minimize latency and maximize security. Which option should they choose?

- A. Point-to-Site VPN
- B. Site-to-Site VPN
- C. ExpressRoute
- D. Azure Application Gateway

Right option: C. ExpressRoute

Explanation: ExpressRoute provides a private connection between on-premises networks and Azure. It is typically chosen when lower latency and stronger network isolation are required than an internet VPN.

Remember: Private dedicated Azure connectivity = ExpressRoute.

Q32. As part of a hybrid cloud deployment at Contoso, you need to connect Contoso's on-premises datacenter to Azure. The solution you choose should minimize expense during the low-scale pilot deployment. Which option will you choose?

- A. Point-to-Site VPN
- B. Site-to-Site VPN
- C. ExpressRoute
- D. Azure Application Gateway

Right option: B. Site-to-Site VPN

Explanation: A site-to-site VPN connects an entire on-premises network to an Azure virtual network. It is usually cheaper and faster to start with than ExpressRoute for a pilot.

Remember: Network-to-network VPN = Site-to-Site.

Q33. For regulatory compliance, you need to ensure Contoso's corporate web apps use TLS 1.2 for encryption. Does Azure App Service support enforcing this specific requirement?

- A. YES
- B. NO

Right option: A. YES

Explanation: Azure App Service lets you configure the minimum TLS version for hosted apps. That supports compliance requirements such as enforcing TLS 1.2.

Remember: TLS minimum version can be configured in App Service.

Q34. Contoso wants to migrate a legacy application to Azure that requires an file share accessible on a UNC path. You need to provide an SMB file share and secure access. The solution should minimize administrative effort. Which service will you use?

- A. OneDrive
- B. SharePoint
- C. Azure VM with a file share
- D. Azure Files

Right option: D. Azure Files

Explanation: Azure Files provides managed SMB file shares that can be accessed with UNC paths. It gives the legacy app a file-share experience without maintaining a file server VM.

Remember: SMB/UNC file share in Azure = Azure Files.

Q35. Storage for Azure VMs is hosted in which Azure Storage type?

- A. File
- B. Table
- C. Disk
- D. Blob

Right option: C. Disk

Explanation: Azure VM operating system and data disks are stored as Azure managed disks. In the quiz choices, this corresponds to Disk storage.

Remember: VM OS and data disks use managed disks.

Q36. Contoso IT wants to configure separate Azure subscriptions for different environments (production, development, test) and products. Can they associate multiple Azure subscriptions to the same Entra ID tenant?

- A. YES
- B. NO

Right option: A. YES

Explanation: An Entra ID tenant can be associated with multiple Azure subscriptions. This lets the same identity directory manage access across separate environments or products.

Remember: One tenant can manage multiple subscriptions.

Q37. Contoso Financial Services regularly creates documents containing sensitive customer data, including personally identifiable information. You need to ensure access is restricted to finance personnel, regardless of where the document travels. Which service should you use to classify and protect these documents?

- A. Microsoft Defender for Cloud Apps
- B. Microsoft Defender for Endpoint
- C. Microsoft Purview
- D. Entra ID Conditional Access

Right option: C. Microsoft Purview

Explanation: Microsoft Purview provides data classification, sensitivity labels, and information protection. It is used to protect sensitive documents even as they move outside their original location.

Remember: Classification, labels, and data protection = Microsoft Purview.

Q38. An Azure _____ is a set of datacenters deployed within a latency-defined perimeter and connected through a dedicated regional low-latency network.

- A. Availability Zone
- B. Region
- C. Geography
- D. Datacenter

Right option: B. Region

Explanation: An Azure region is a set of datacenters within a latency-defined perimeter connected by a regional low-latency network. Availability zones are smaller fault-isolated locations inside a region.

Remember: Region is the broad datacenter geography.

Q39. Which Azure storage redundancy option does Microsoft recommend for maximum redundancy and recoverability?

- A. Locally Redundant Storage (LRS)
- B. Geo-Redundant Storage (GRS)
- C. Zone Redundant Storage (ZRS)
- D. Geo Zone Redundant Storage (GZRS)

Right option: D. Geo Zone Redundant Storage (GZRS)

Explanation: GZRS combines zone redundancy within a region with geo-replication to a paired region. That gives stronger redundancy and recoverability than LRS, ZRS, or GRS alone.

Remember: GZRS protects across zones and a paired region.

Q40. An _____ protects against VM failures due to failures in updates, power, or network connectivity.

- A. Availability Set
- B. Availability Zone
- C. VM scale set
- D. Azure VM

Right option: A. Availability Set

Explanation: An availability set spreads VMs across fault domains and update domains. This reduces impact from hardware, power, network, or planned maintenance failures.

Remember: Availability sets spread VMs across fault and update domains.

Q41. You need to manage your Azure VMs using the Azure portal. Which URL would you use to manage the Azure VMs?

- A. <https://portal.azurewebsites.net>
- B. <https://portal.microsoft.com>
- C. <https://portal.azure.microsoft.com>
- D. <https://portal.azure.com>

Right option: D. <https://portal.azure.com>

Explanation: The Azure portal for managing Azure resources is portal.azure.com. Other Microsoft portals exist, but this is the main Azure management portal.

Remember: The Azure portal URL is portal.azure.com.

Q42. Contoso IT wants to develop modern application components using a serverless architecture. Which Azure service is best for hosting code as part of a services infrastructure?

- A. Azure Logic Apps
- B. Azure Functions
- C. Azure Service Bus
- D. Azure Automation

Right option: B. Azure Functions

Explanation: Azure Functions is the Azure serverless compute service for hosting and running code. Logic Apps is more focused on visual workflows and integrations.

Remember: Serverless code execution = Azure Functions.

Q43. You can monitor health and performance of microservices applications running on Azure Kubernetes Service (AKS) with:

- A. Azure Monitor
- B. Azure App Insights
- C. Azure Sentinel
- D. Microsoft Defender for Cloud

Right option: B. Azure App Insights

Explanation: Application Insights is the application performance monitoring feature used for requests, dependencies, failures, and availability. It is the app-focused monitoring tool for microservices.

Remember: Application telemetry = Application Insights.

Q44. You need to store unstructured data, such as images, video files, social media posts on Azure Storage. Which type of Azure Storage is optimized for storage of large amounts of unstructured data?

- A. Table Storage
- B. Queue Storage
- C. Disk Storage
- D. Blob Storage

Right option: D. Blob Storage

Explanation: Blob Storage is optimized for large amounts of unstructured object data such as images, video, backups, and documents. Table and Queue Storage are for structured key/value data and messages.

Remember: Unstructured object data = Blob Storage.

Q45. Contoso IT has deployed a group of Azure VMs for hosting production. They want best practices recommendations for high availability of their VM resources. Will the Azure Advisor tool provide recommendations for these existing resources?

- A. YES
- B. NO

Right option: A. YES

Explanation: Azure Advisor analyzes deployed resources and provides best-practice recommendations. Its reliability recommendations can include high-availability improvements for existing VMs.

Remember: Best-practice recommendations = Azure Advisor.

Q46. The Contoso IT Operations team needs to monitor their customer-facing web apps for performance anomalies. Which Azure service will best fulfill this need?

- A. Azure WebJobs
- B. Azure Application Insights
- C. Azure Automation
- D. Azure Sentinel

Right option: B. Azure Application Insights

Explanation: Application Insights monitors application performance and detects anomalies in web applications. It is more specific to app telemetry than Azure Automation or Sentinel.

Remember: Web app performance telemetry = Application Insights.

Q47. You need to automate responses to some alerts from Azure Monitor. Which service supports automating responses and corrective actions in this scenario?

- A. Power Automate
- B. Azure Logic Apps
- C. Azure Web Jobs
- D. Azure Automation

Right option: D. Azure Automation

Explanation: Azure Automation can run runbooks and operational scripts as responses to alerts. This makes it suitable for corrective actions triggered from monitoring.

Remember: Operational runbooks and corrective scripts = Azure Automation.

Q48. The Contoso IT Operations team needs to aggregate events from a large number of resources hosted in Azure for correlation, alerting, and reporting. Which Azure service would you use to centrally collect, store, and act on events?

- A. Azure App Insights
- B. Azure Monitor
- C. Azure Data Lake
- D. Azure Event Hub

Right option: B. Azure Monitor

Explanation: Azure Monitor is the central Azure service for collecting, storing, analyzing, alerting on, and acting on metrics and logs from Azure resources.

Remember: Central metrics, logs, and alerts platform = Azure Monitor.

Q49. You need to alert on service failures in your Azure services, such as web app instances hosted in App Service, and Azure VMs that stop running for any reason. Which tool should you use?

- A. Azure App Insights
- B. Azure Log Analytics
- C. Azure Monitor
- D. Microsoft Defender for Cloud

Right option: C. Azure Monitor

Explanation: Azure Monitor supports alerts based on metrics and logs, including failures or stopped resources. It is the best fit for alerting across Azure services and VMs.

Remember: Alerts for Azure resource conditions = Azure Monitor.

Q50. You can find recommendations for security best practices security configuration for Azure Kubernetes Service (AKS) with:

- A. Azure Monitor
- B. Azure App Insights
- C. Microsoft Sentinel
- D. Microsoft Defender for Cloud

Right option: D. Microsoft Defender for Cloud

Explanation: Microsoft Defender for Cloud evaluates cloud security posture and provides security recommendations. It includes recommendations for services such as AKS.

Remember: Cloud security posture = Defender for Cloud.

Q51. Azure VMs in different virtual networks can communicate by default

- A. TRUE
- B. FALSE

Right option: B. FALSE

Explanation: Virtual networks are isolated by default. VMs in different VNets need peering, VPN, ExpressRoute, or other routing to communicate.

Remember: Different VNets are isolated unless connected.

Q52. Which Microsoft solution provides support for passwordless authentication on Windows 10 and 11 systems?

- A. Entra ID Conditional Access
- B. Microsoft Authenticator
- C. Windows Hello for Business
- D. Microsoft Purview

Right option: C. Windows Hello for Business

Explanation: Windows Hello for Business is Microsoft's passwordless sign-in solution for Windows devices. It uses strong authentication methods such as biometrics or PIN-backed credentials.

Remember: Passwordless Windows sign-in = Windows Hello for Business.

Q53. You have deployed an Azure VM hosting a line-of-business web application. You need to provide access to the application over the Internet via HTTP/S. You add a security rule to the Network Security Group (NSG) to allow inbound traffic from the Internet. Does this solution meet the requirement?

- A. YES
- B. NO

Right option: A. YES

Explanation: A Network Security Group controls inbound and outbound network traffic to Azure resources. Allowing inbound HTTP/HTTPS permits internet web traffic to reach the VM when the app and public routing are configured.

Remember: NSGs allow or deny inbound and outbound traffic.

Q54. Contoso IT has deployed multiple Azure VM across 15 virtual networks. How can you most securely limit inbound traffic and protect these VMs from unwanted inbound requests? (choose the best answer)

- A. Create a network security group (NSG)
- B. Deploy an Azure Firewall
- C. Deploy an Azure Load Balancer
- D. Deploy a virtual network gateway

Right option: B. Deploy an Azure Firewall

Explanation: Azure Firewall provides centralized, stateful network filtering and protection across many virtual networks. It is stronger for broad inbound traffic control than managing many separate NSGs alone.

Remember: Central network filtering = Azure Firewall.

Q55. You need to define and enforce corporate standards for new and existing Azure resource deployments in all of Contoso's Azure subscriptions. What should you use to meet this objective?

- A. Azure Policy
- B. Microsoft Defender for Cloud
- C. Microsoft Sentinel
- D. Azure Advisor

Right option: A. Azure Policy

Explanation: Azure Policy defines rules for allowed or required resource configurations. It can audit, deny, or remediate resources that do not match corporate standards.

Remember: Enforce resource rules = Azure Policy.

Q56. Tailspin Toys relies heavily on Entra ID for cloud identity. They want to more effectively protect their identities from external threats. Which service should they choose?

- A. Azure Identity Protection
- B. Microsoft Defender for Endpoint
- C. Azure DDoS
- D. Microsoft Information Protection

Right option: A. Azure Identity Protection

Explanation: Entra ID Identity Protection detects identity-based risks such as risky users and risky sign-ins. It helps protect cloud identities from compromise.

Remember: Risky users and sign-ins = Identity Protection.

Q57. In a Site-to-Site VPN, the___ is the cross-premises gateway that connects your Azure Virtual Network with your on-premises VPN appliances

- A. Azure Application Gateway
- B. Azure Virtual Network Gateway
- C. Local Network Gateway
- D. Private Endpoint

Right option: B. Azure Virtual Network Gateway

Explanation: The Azure Virtual Network Gateway is the Azure-side VPN gateway. It connects an Azure virtual network to on-premises VPN devices in a site-to-site VPN.

Remember: Azure VPN gateway service = Virtual Network Gateway.

Q58. You need to prevent accidental deletion of Azure resources in your subscription. Which feature will meet this requirement? (choose the best answer)

- A. RBAC
- B. Resource Locks
- C. Security groups
- D. Azure policies

Right option: B. Resource Locks

Explanation: Resource locks protect Azure resources from accidental deletion or modification. A delete lock is the common choice when the goal is preventing accidental deletion.

Remember: Prevent deletion = Resource locks.

Q59. You are responsible for creating Azure resources at Contoso using ARM templates. You need to ensure Azure resources are only created in approved regions. What should you use to enforce this requirement

- A. Azure locks
- B. Azure Policy
- C. Azure Blueprint
- D. Microsoft Defender for Cloud

Right option: B. Azure Policy

Explanation: Azure Policy can enforce allowed locations so resources are created only in approved regions. Locks protect existing resources but do not define regional compliance rules.

Remember: Allowed regions = Azure Policy.

Q60. Contoso IT assigns permissions at the resource group level for all resources deployed to Azure. Will resources in the resource group inherit permissions assigned to the resource group?

- A. YES
- B. NO

Right option: A. YES

Explanation: RBAC assignments are inherited down the Azure scope hierarchy. Permissions assigned at a resource group apply to resources inside that group unless overridden by scope design.

Remember: RBAC inheritance flows down the scope hierarchy.

Q61. Contoso has implemented a hybrid, synchronized identity model, consisting of on-premises Active Directory and Entra ID. They want to more effectively protect their on-premises identities from external threats. Which service should they choose?

- A. Azure Identity Protection
- B. Microsoft Defender for Endpoint
- C. Azure DDoS
- D. Microsoft Information Protection

Right option: B. Microsoft Defender for Endpoint

Explanation: The quiz maps this to Defender for Endpoint, which protects devices and servers from threats. For identity-specific attacks against on-premises Active Directory, remember that Microsoft Defender for Identity is the dedicated product when it appears as an option.

Remember: Endpoint and device threat protection = Defender for Endpoint.

Q62. You need to implement multi-factor authentication for your Entra ID users. However, you only want to prompt for an additional authentication factor when users are not in a trusted location on an unmanaged device. Which feature should you implement? (choose the best answer)

- A. Microsoft Purview
- B. Identity Protection
- C. Conditional Access
- D. Privileged Identity Management

Right option: C. Conditional Access

Explanation: Conditional Access applies access rules based on signals such as location, device state, and risk. It can require MFA only when the sign-in is outside trusted conditions.

Remember: Conditional Access applies rules based on signals.

Q63. The Contoso Security team has implemented a new security policy. When users connect from an outside corporate offices, they must be prompted for MFA. Which feature will you implement?

- A. Entra ID Privileged Identity Management
- B. One Time Passwords (OTP)
- C. Entra ID Identity Protection
- D. Entra ID Conditional Access

Right option: D. Entra ID Conditional Access

Explanation: Conditional Access is the policy engine for decisions such as requiring MFA when users sign in from outside corporate offices. OTP is a factor, not the policy engine.

Remember: Outside-office MFA rule = Conditional Access.

Q64. You need to support OATH tokens (one-time password) as a second authentication factor for Entra ID. What Microsoft solution enables use of OATH tokens for Entra ID and other identity providers?

- A. Entra ID Identity Protection
- B. Microsoft Authenticator
- C. Conditional Access
- D. Entra ID Multi-Factor Authentication

Right option: B. Microsoft Authenticator

Explanation: Microsoft Authenticator provides one-time passcodes and other second-factor methods for Entra ID and other identity providers. That makes it the best answer for OATH-style second-factor support in this quiz.

Remember: MFA app and passcodes = Microsoft Authenticator.

Q65. The Contoso Legal Department has asked Contoso IT to verify whether the Contoso's Azure environment meets regulatory requirements. Which service should you use to answer this question?

- A. Azure Advisor
- B. Microsoft Defender for Cloud
- C. Microsoft Purview
- D. Azure Policy

Right option: B. Microsoft Defender for Cloud

Explanation: Microsoft Defender for Cloud includes security posture and regulatory compliance views. It can help assess whether Azure resources align with compliance requirements.

Remember: Security posture and compliance = Defender for Cloud.

Q66. What are the three foundational principles of Zero Trust?

- A. 1) Verify explicitly 2) Use least privilege access 3) Assume breach
- B. 1) Verify explicitly 2) Use defense in depth 3) Assume breach
- C. 1) Verify explicitly 2) Trust but verify 3) Assume breach
- D. 1) Verify explicitly 2) Use least privilege access 3) Trust must be earned

Right option: A. 1) Verify explicitly 2) Use least privilege access 3) Assume breach

Explanation: Zero Trust is based on verifying explicitly, using least privilege access, and assuming breach. These principles reduce implicit trust in users, networks, and devices.

Remember: Zero Trust: verify, least privilege, assume breach.

Q67. Contoso Electronics is a global retailer. The Contoso Cloud Architecture team needs to simplify deployments of new environments in Azure, including Azure Resource Manager (ARM) templates, role-based access, and policies. Which Azure service enables delivery of templates for repeatable deployment and configuration of new subscriptions and environments? (choose the best answer)

- A. Azure Policy initiatives
- B. Azure Policy
- C. ARM templates
- D. Azure Blueprints

Right option: D. Azure Blueprints

Explanation: Azure Blueprints packaged repeatable environment setup, including ARM templates, role assignments, and policies. ARM templates alone deploy resources, but Blueprints added governance around the deployment.

Remember: Governance plus deployment package = Blueprints.

Q68. Your company is planning to host services in Azure. You want to leverage identities in Entra ID, but still need to support on-premises identities in Active Directory. Can you support single sign-on (SSO) and multi-factor authentication for both on-premises and cloud with Entra ID?

- A. YES
- B. NO

Right option: A. YES

Explanation: Entra ID can support hybrid identity with on-premises Active Directory through synchronization or federation. This enables SSO and MFA for both cloud and hybrid identity scenarios.

Remember: Hybrid identity connects AD DS with Entra ID.

Q69. You need to configure access to Office 365 resources for users at Contoso. To group users for assignment of permissions, which of the following options would you use?

- A. Microsoft 365 group
- B. Resource group
- C. Security group
- D. Management group

Right option: C. Security group

Explanation: Security groups are used to group users for access and permission assignment. Microsoft 365 groups are mainly collaboration groups with shared workspace resources.

Remember: Permission assignment groups = Security groups.

Q70. Implementing Azure MFA (multi-factor authentication) will ensure we know who the user is who they claim to be. This is an example of:

- A. Authorization
- B. Authentication
- C. Integrity
- D. Confidentiality

Right option: B. Authentication

Explanation: MFA helps prove that the user is who they claim to be. Proving identity is authentication; deciding what they can access is authorization.

Remember: MFA proves identity: authentication.

Q71. You need to ensure on-premises file shares and Azure Files remain in sync in both directions, regardless of where the file was changed. Which tool or service would you choose? (choose the best answer)

- A. AzCopy
- B. Azure Storage Explorer
- C. Azure File Sync
- D. Azure Data Box

Right option: C. Azure File Sync

Explanation: Azure File Sync synchronizes on-premises Windows file shares with Azure Files. It supports keeping file content in sync across locations.

Remember: Sync on-premises file servers with Azure Files = Azure File Sync.

Q72. Contoso's Security team wants to implement selective use of multi-factor authentication (MFA) based on multiple factors related to the authentication request, such as device health and sign-in risk. Which services should they implement?

- A. Conditional Access and Identity Protection
- B. Identity Protection and Entitlement Management
- C. Multi-Factor Authentication (MFA) and Identity Protection
- D. Windows Hello for Business and Conditional Access

Right option: A. Conditional Access and Identity Protection

Explanation: Identity Protection detects sign-in and user risk. Conditional Access uses those signals and other conditions to require MFA, block access, or allow access.

Remember: Risk signals plus access rules = Identity Protection + Conditional Access.

Q73. Contoso has deployed resources across multiple Azure regions for multiple business units. They have a requirement to generate cost and chargeback reporting to track the Azure costs to be charged back to each business unit. Which Azure feature should they use to simplify this task?

- A. Tags
- B. Resource Locks
- C. Resource Groups
- D. Management Groups

Right option: A. Tags

Explanation: Tags are metadata labels placed on Azure resources, such as department, app, or cost center. They make cost reporting and chargeback easier.

Remember: Department and cost metadata labels = Tags.

Q74. You need to identify and enforce Contoso's corporate standards across new and existing Azure deployments. Which service would you choose to achieve this requirement? (choose the best answer)

- A. Azure Blueprints
- B. Azure Policy
- C. Azure Resource Manager (ARM) templates
- D. Azure Automation

Right option: B. Azure Policy

Explanation: Azure Policy is used to define, audit, and enforce standards across new and existing Azure resources. It is the main governance tool for compliance rules.

Remember: Governance standards, audit, and enforcement = Azure Policy.

Q75. Using role based access control (RBAC) in Azure, we can determine which resources and services a user has access to. This is an example of:

- A. Authorization
- B. Authentication
- C. Integrity
- D. Confidentiality

Right option: A. Authorization

Explanation: RBAC determines what actions a user can perform on which resources. That is authorization because it controls permitted access after the user is identified.

Remember: RBAC is authorization, not authentication.

Q76. Azure China and Azure Government are examples of:

- A. Private clouds
- B. Sovereign regions
- C. Azure geographies
- D. Management groups

Right option: B. Sovereign regions

Explanation: Azure China and Azure Government are sovereign cloud/region offerings designed for specific legal, regulatory, or national requirements. They are not private clouds owned by one customer.

Remember: Government and China clouds are sovereign regions.

Q77. You need to ensure no one (including administrators) can create additional resources in a Azure resource group. What will you do to achieve this objective?

- A. Azure Policy
- B. Role based access control (RBAC)
- C. Resource locks
- D. Microsoft Defender for Cloud

Right option: C. Resource locks

Explanation: A read-only resource lock on a resource group prevents changes, including creating additional resources, even for users with high permissions. RBAC controls permissions, but locks add a protection layer.

Remember: Read-only locks can block changes.

Q78. Which of the following describes authentication?

- A. Determines which resources you can access
- B. Defines the services and regions you can access
- C. Validates that a user is who they claim to be
- D. Establishes your resource access

Right option: C. Validates that a user is who they claim to be

Explanation: Authentication validates identity: it answers 'Who are you?' Authorization comes afterward and answers 'What are you allowed to access?'

Remember: Authentication asks: Who are you?

Q79. You need to track resource consumption by application and department for cost tracking and chargeback. Which of the following will enable this capability.

- A. Azure Monitor
- B. Tags
- C. Azure
- D. Management Groups

Right option: B. Tags

Explanation: Tags let you label resources by application, department, owner, or cost center. Cost reports can then group consumption using those labels.

Remember: Cost grouping labels = Tags.

Q80. You need to automate movement of multiple files to a storage account. What command line tool can you use to script copying blobs or files to or from your storage account? (choose the best answer)

- A. Azure File Sync
- B. Azure Storage Explorer
- C. AzCopy
- D. Azure Data Box

Right option: C. AzCopy

Explanation: AzCopy is a command-line tool for copying blobs and files to or from Azure Storage. It is scriptable, which fits automated file movement.

Remember: Scripted storage copy = AzCopy.

Q81. Which Azure storage redundancy option would you select for development workloads where minimizing expense is the highest priority?

- A. Zone Redundant Storage (ZRS)
- B. Geo-Redundant Storage (GRS)
- C. Geo-Zone Redundant Storage (GZRS)
- D. Locally Redundant Storage (LRS)

Right option: D. Locally Redundant Storage (LRS)

Explanation: LRS keeps copies within a single region/location and is the lowest-cost redundancy option. It is suitable for low-priority development workloads where cost matters most.

Remember: Lowest redundancy cost = LRS.

Q82. Contoso needs Azure capacity to support spikes in request traffic to their load-balanced web farm during the holiday shopping season. A consultant recommends deploying the website to a virtual machine scale set in Azure. Will this meet the requirement?

- A. YES
- B. NO

Right option: A. YES

Explanation: Virtual machine scale sets can increase or decrease VM instances behind a load-balanced application. This helps handle seasonal or temporary traffic spikes.

Remember: Scale sets add or remove VM instances.

Q83. Which type of Azure storage stores NoSQL data in Azure, including a schemaless key/attribute store?

- A. Queue Storage
- B. Table Storage
- C. Blob Storage
- D. File Storage

Right option: B. Table Storage

Explanation: Table Storage is a NoSQL key/attribute store with a schemaless design. Queue Storage is for messages, not general NoSQL entity data.

Remember: NoSQL key/attribute entities = Table Storage.

Q84. You need to identify deviations from Microsoft security best practices in your Azure cloud infrastructure. Which service should you use?

- A. Azure Advisor
- B. Microsoft Defender for Cloud
- C. Azure Monitor
- D. Azure Key Vault

Right option: B. Microsoft Defender for Cloud

Explanation: Microsoft Defender for Cloud compares your cloud environment against security best practices and surfaces recommendations. It is used for cloud security posture management.

Remember: Security recommendations = Defender for Cloud.

Q85. Contoso IT is planning to migrate all on-premises data to Azure. The Legal Department has asked for verification that Azure complies with Contoso's regulatory obligations, such as HIPAA and PCI DSS. Which Azure service can be used to monitor for regulatory compliance?

- A. Azure App Insights
- B. Azure Advisor
- C. Azure Monitor
- D. Microsoft Defender for Cloud

Right option: D. Microsoft Defender for Cloud

Explanation: Defender for Cloud includes a regulatory compliance dashboard and security assessments. It helps monitor whether cloud resources align with standards such as HIPAA or PCI DSS.

Remember: Regulatory compliance dashboard = Defender for Cloud.

Q86. You need to calculate the estimated cost of a set of Azure resources before you deploy them. Which service or tool will you use?

- A. Azure Advisor
- B. Azure Pricing Calculator
- C. Azure Cost Management
- D. Azure TCO Calculator

Right option: B. Azure Pricing Calculator

Explanation: The Azure Pricing Calculator estimates the cost of resources before deployment. Cost Management is mainly for analyzing actual spend after resources exist.

Remember: Pre-deployment cost estimate = Pricing Calculator.

Q87. As part of a migration of on-premises VM workloads to Azure VMs, Contoso wants to minimize costs. Since the VMs will be running in Azure for an extended period of time (years), what option should they consider to reduce hosting costs?

- A. Azure Cost Management
- B. Azure Reservations
- C. Azure Advisor
- D. VM Scale Sets

Right option: B. Azure Reservations

Explanation: Azure Reservations reduce cost for resources that will run for a long time. They are a better fit than VM Scale Sets when the main goal is price reduction for predictable VM usage.

Remember: Long-running resource discount = Reservations.

Q88. A _____ is a virtual network gateway that sends encrypted traffic between an Azure VNET and an on-premises location over the Internet

- A. App gateway
- B. Web proxy
- C. VPN gateway
- D. Private endpoint

Right option: C. VPN gateway

Explanation: A VPN gateway sends encrypted traffic between an Azure virtual network and an on-premises network over the internet. Application Gateway is for web traffic routing, not site connectivity.

Remember: Encrypted on-premises-to-Azure tunnel = VPN Gateway.

Q89. Contoso IT Operations has been tasked with providing recommendations on how to reducing cost of running Azure VMs. Which service should they use to gather recommendations?

- A. Microsoft Defender for Cloud
- B. Azure Advisor
- C. Azure Monitor
- D. Azure App Insights

Right option: B. Azure Advisor

Explanation: Azure Advisor provides recommendations to optimize Azure resources, including cost recommendations for VMs. It is the recommendation engine; Cost Management is for spend analysis.

Remember: Optimization recommendations = Azure Advisor.

Q90. Contoso is planning to move several on-premises services to Azure PaaS and IaaS solutions. The CIO has asked for a tool to estimate the cost of hosting these resources in Azure. Tom suggests Azure Cost Management to estimate costs prior to migration. Does Tom's solution meet the solution criteria?

- A. YES
- B. NO

Right option: B. NO

Explanation: Azure Cost Management analyzes and manages costs after resources are running. For estimating before migration, the Pricing Calculator or TCO Calculator is the appropriate tool.

Remember: Cost Management analyzes actual spend, not estimates.

Q91. To manage service lifecycle with more granularity, Contoso cloud architects have designed a model that involves a large number of resource groups. Will Contoso incur additional costs for the resource groups?

- A. YES
- B. NO

Right option: B. NO

Explanation: Resource groups are logical containers and do not have a direct charge. You pay for the resources inside them, not for creating many resource groups.

Remember: Resource groups are free containers.

Q92. You plan to deploy several Azure VMs. The applications running on these VMs should remain available if a single Azure datacenter fails. You opt to deploy VMs to multiple availability zones. Does your solution meet the solution criteria?

- A. YES
- B. NO

Right option: A. YES

Explanation: Availability zones are separate datacenter locations within a region. Deploying VMs across zones helps keep applications available if one datacenter fails.

Remember: Availability zones improve datacenter-level resilience.

Q93. Contoso IT has deployed a group of VMs in Azure. They want to identify recommendations on how to reduce the cost of running these VMs. Which tool should they use? (choose the best answer)

- A. Azure Price Calculator
- B. Azure Advisor
- C. Azure Cost Management
- D. Microsoft Defender for Cloud

Right option: B. Azure Advisor

Explanation: Azure Advisor analyzes deployed resources and recommends optimizations. For VMs, it can identify cost-saving opportunities such as resizing or shutting down underused resources.

Remember: Cost-saving recommendations = Azure Advisor.

Q94. Azure App Service supports which of the following application type(s)?

- A. Web apps
- B. Mobile apps
- C. API apps
- D. All the above

Right option: D. All the above

Explanation: Azure App Service supports several application types, including web apps, mobile app back ends, and API apps. Therefore all listed types are supported.

Remember: App Service hosts web, mobile, and API apps.

Q95. Which virtual machine configuration supports a more resilient, highly available environment by staggering VM updates and ensuring varied power and network connectivity?

- A. Virtual machine scale sets
- B. Virtual machine availability sets
- C. virtual cluster
- D. Azure Virtual Desktop

Right option: B. Virtual machine availability sets

Explanation: Availability sets place VMs across fault domains and update domains. This staggers maintenance and reduces shared power/network/hardware failure risk.

Remember: Availability sets spread VMs for maintenance and fault tolerance.

Q96. Tailspin Toys uses a pay-as-you-go (PAYG) subscription in Azure. PAYG generally provides lowest cost over time, but least flexibility in terms of shifting consumption to new services.

- A. TRUE
- B. FALSE

Right option: B. FALSE

Explanation: Pay-as-you-go provides high flexibility because you can change or stop consumption, but it is not generally the lowest long-term cost. Reservations are usually cheaper for steady usage.

Remember: Pay-as-you-go is flexible, not cheapest long-term.

Q97. Azure Container Instances enable running containers without host servers to manage.

- A. YES
- B. NO

Right option: A. YES

Explanation: Azure Container Instances runs containers without requiring you to manage host VMs or container orchestration infrastructure. It is a serverless-style container option.

Remember: Run containers without managing VMs = ACI.

Q98. Azure Container Instances enable elastic bursting for Azure Kubernetes Service

- A. YES
- B. NO

Right option: A. YES

Explanation: Azure Container Instances can be used with AKS virtual nodes for burst capacity. This lets AKS handle temporary spikes without pre-provisioning extra cluster nodes.

Remember: AKS virtual nodes use ACI for bursts.

Q99. Azure VMs in different subnets in the same virtual network can communicate by default

- A. YES
- B. NO

Right option: A. YES

Explanation: Subnets inside the same virtual network have default system routes that allow communication. NSGs, firewalls, or custom routes can block it, but it is allowed by default.

Remember: Same VNet subnets communicate by default.

Q100. Azure DDoS, which protects your Azure resources against distributed denial of services attacks, includes both a Basic and Standard tiers.

- A. YES
- B. NO

Right option: A. YES

Explanation: At the level tested by this quiz, Azure DDoS includes baseline protection and a higher paid tier with enhanced protection and visibility. So the statement about Basic and Standard tiers is marked true.

Remember: DDoS has baseline and enhanced protection tiers.
